`jester` v0.2.0: Improved inference of the equation of state with multimessenger data

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Table of Contents

1. Introduction
2. Results
3. Conclusion

Contents

1 Introduction

2 Results

3 Conclusion

Motivation

Contents

1 Introduction

2 Results

3 Conclusion

Sampler systematics

Analyze GW170817

EOS model	SMC	NS
MM+CSE	$12.60^{+1.04}_{-1.42}$	$12.47^{+1.14}_{-1.37}$
Spectral	$12.49^{+1.01}_{-1.33}$	$12.40^{+1.04}_{-1.35}$

Table 1: $R_{1.4}$ [km] at 90% credibility (with radio).

Contents

1 Introduction

2 Results

3 Conclusion

Conclusion

- PE often uses agnostic priors, missing valuable information
- **Neural priors**: flexible way to encode non-trivial prior information
- Case study: neutron star physics from
  - Population models
  - EOS constraints
- Two highlights:
  1. **Bayesian source classification** (BNS vs. NSBH)
  2. **Informed parameter constraints** (narrower posteriors)
- Implemented in BILBY
- Data-driven approach: easy to extend/generalize

Thanks for listening!

References I